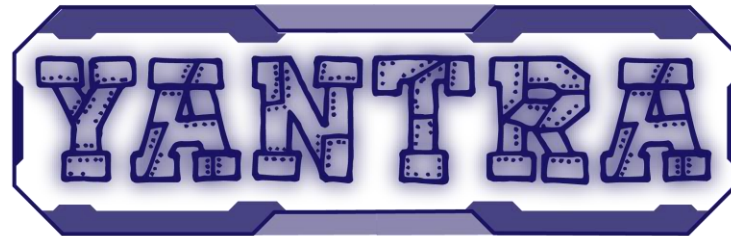
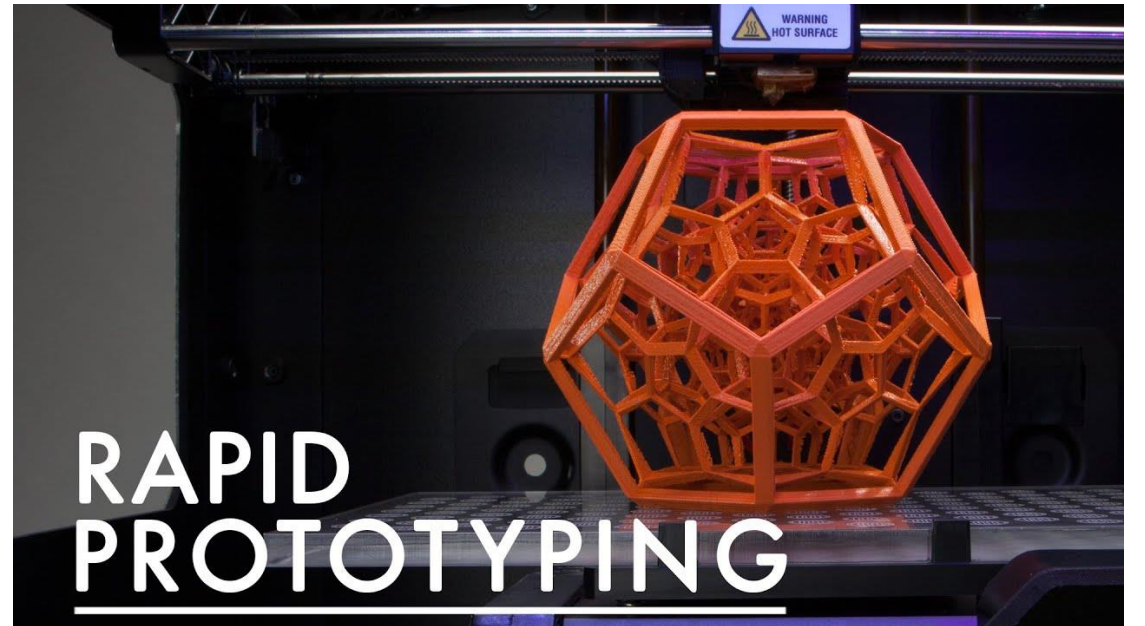
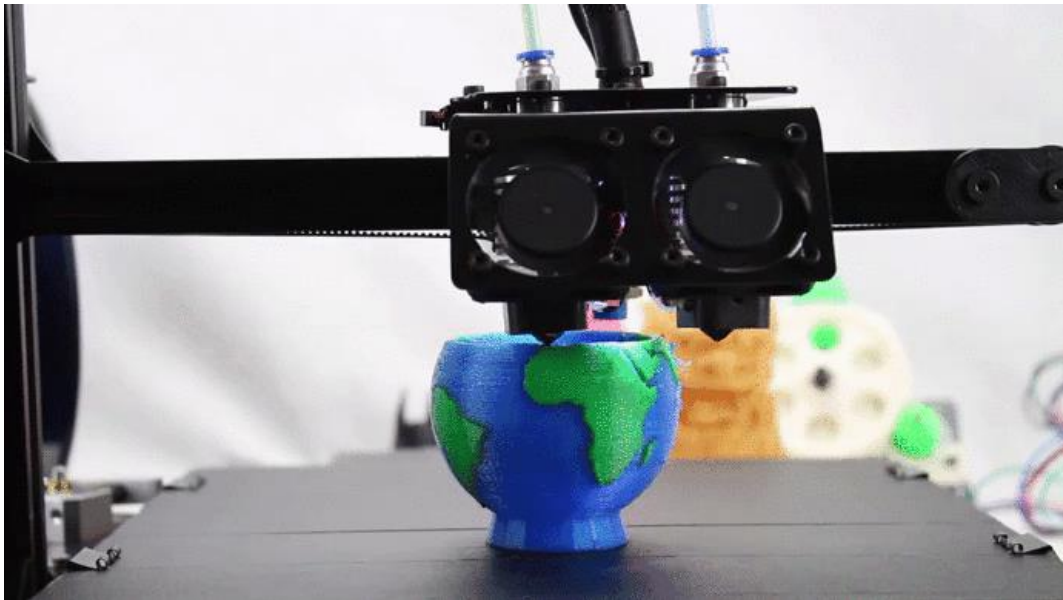
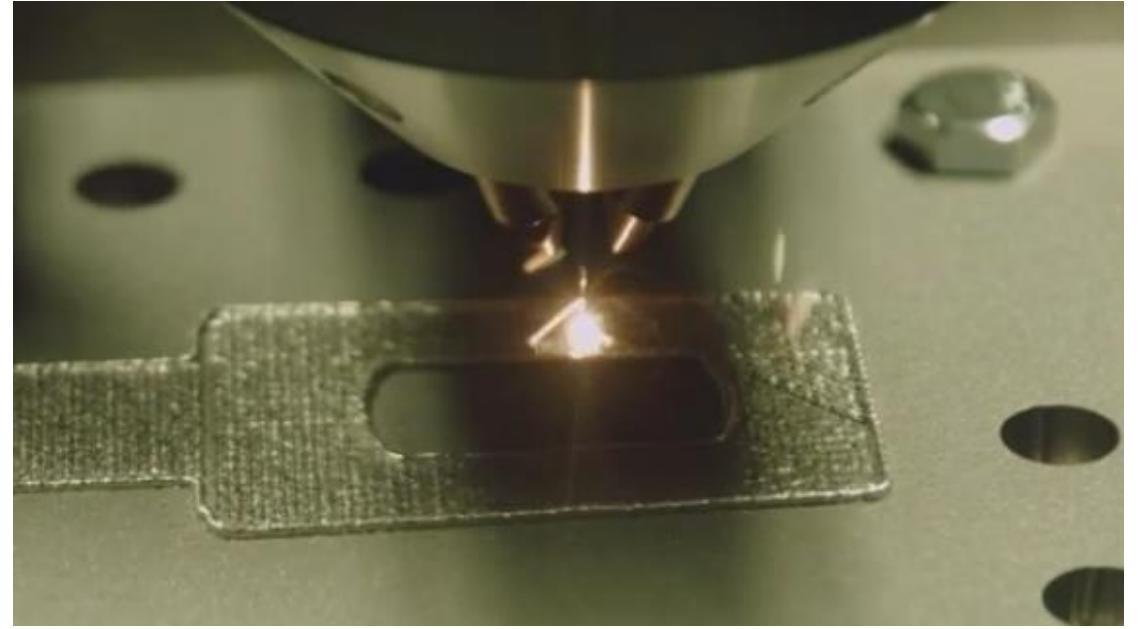
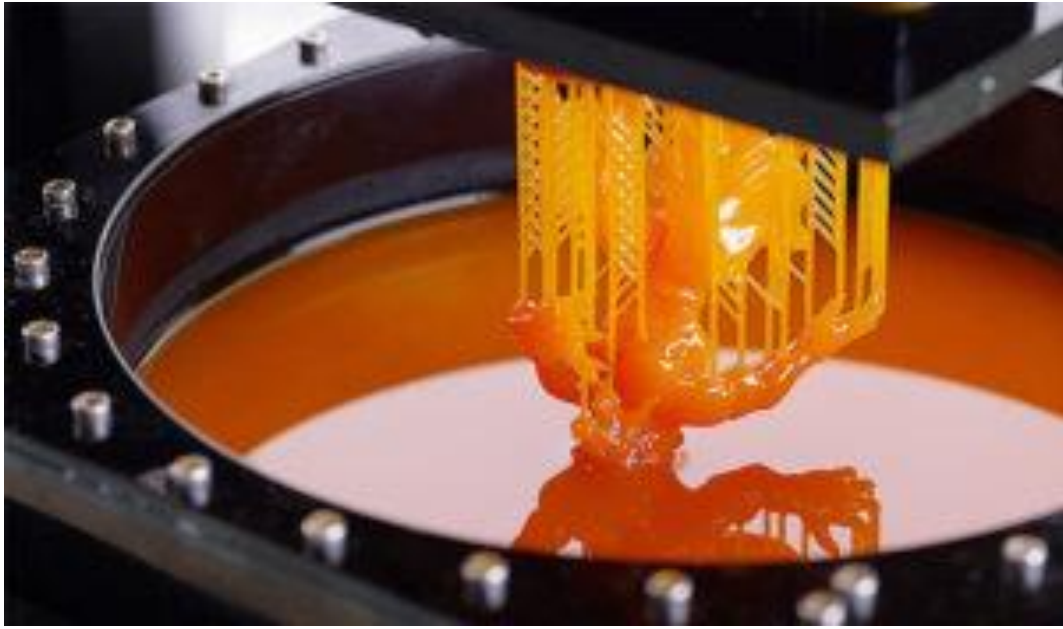


ADDITIVE MANUFACTURING





Introduction

- Additive manufacturing is the formalized term for what used to be called rapid prototyping and what is popularly called **3D Printing**.
- The term **rapid prototyping** (RP) is used in a variety of industries to describe a process for rapidly creating a system or part representation before final release or commercialization.
- In other words, the emphasis is on creating something quickly and that the output is a prototype or basis model from which further models and eventually the final product will be derived

Introduction

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- The term rapid prototyping (RP) is used in a variety of industries to describe a process for rapidly creating a system or part representation before final release or commercialization



- The **rapid prototyping** term is inadequate
- and in particular does not effectively describe more recent applications of the technology
- Many parts are in fact now directly manufactured in these machines, so it is not possible for us to label them as “prototypes.”
- recently adopted ASTM consensus standards now use the term **additive manufacturing**

The Generic AM Process

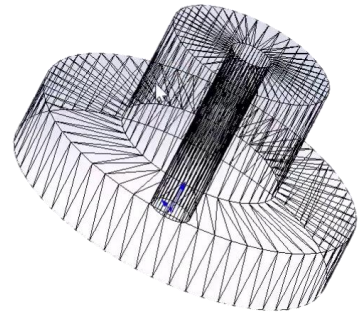
Step 1: CAD

- must start from a software model that fully describes the external geometry.
- use of almost any professional CAD solid modeling software, but the output must be a 3D solid or surface representation



Step 2: Conversion to STL

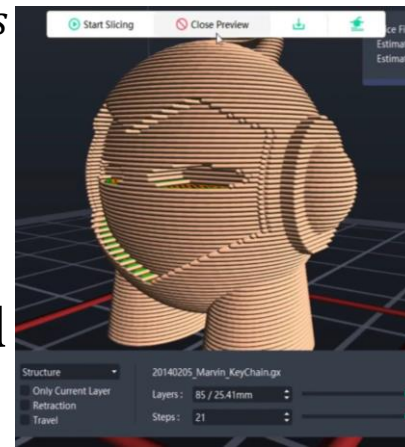
- STL file describes the external closed surfaces of the original CAD model
- forms the basis for calculation of the slices.



STL (Stereolithography) is a file format native to the stereolithography CAD software created by 3D Systems. STL has several after-the-fact backronyms such as "Standard Triangle Language" and "Standard Tessellation Language".

Step 3: Transfer to AM Machine and STL File Manipulation

- general manipulation of the file so that it is the correct size, position, and orientation for building.



Step 4: Machine Setup

- The AM machine must be properly set up prior to the build process. Such settings build parameters like the material constraints, energy source, layer thickness, timings, etc.



Building layer by layer

Step 5: Build

- Building the part is mainly an automated process
- Only superficial monitoring to ensure no errors have taken place like running out of material, power or software glitches, etc.



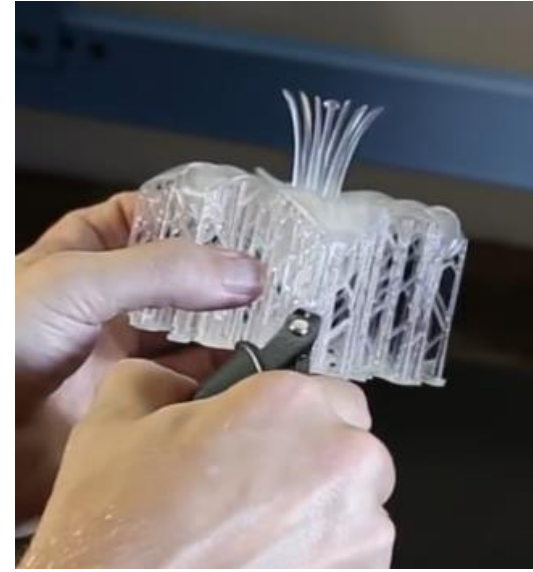
Step 6: Removal

- Once the build has completed, the parts must be removed
- This may require interaction with the machine



Step 7: Post-processing

- parts may require an amount of additional cleaning up
- Parts may be weak at this stage or they may have supporting features that must be removed.
- This therefore often requires time and careful, experienced manual manipulation.



Step 8: Application

- they may require priming and painting to give an acceptable surface texture and finish.
- They may also be required to be assembled together with other mechanical or electronic components to form a final model or product.



Table 1.2: Classification of the GMPs/RP based on the state of raw material

State of material	Type	Mechanism	Energy type	Energy source	Process
Liquid	Photo-polymers	Liquid photo-polymerization	Monochromatic light	Lamp	Solid ground curing (SGC)
				Laser beam	Stereolithography (STL)
				Holography	Holographic interference solid (HIS)
			Light(two frequencies)	Two laser beams	Beam interference solidification (BIS)
	Thermo	Liquid thermal	Heat	Laser beam	Thermal stereolithography (TSTL)
	Non-metals	Melting and solidification	Heat	Heated nozzle	Fused deposition modelling (FDM)
					Ballistic particle manufacturing (BPM)
	Metals	Melting and solidification	Heat	Electric arc	Shape melting
				Laser beam	Fused deposition modelling (FDM)
				Electrochemical discharge	Fused deposition modelling (FDM)

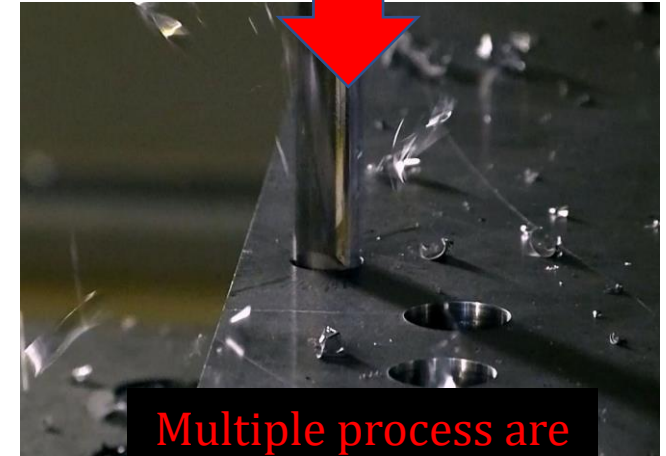
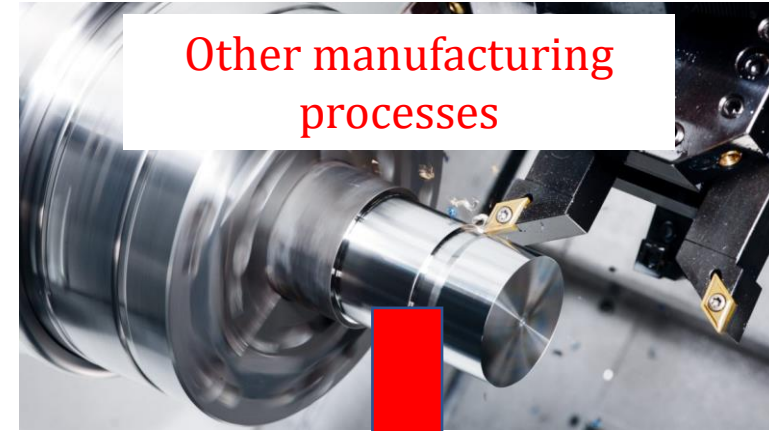
Table 1.2: Classification of the GMPs/RP based on the state of raw material (continued)

State of material	Type	Mechanism	Energy type	Energy source	Process
Solid	Thin sheets and foils	Selective gluing and cutting	Adhesive bonding and	Glue and laser beam	Laminated Object manufacturing (LOM)
	Semi-polymerized plastic foils	Foil polymerization	Light	Lamp	Solid foil polymerization (SFP)
Powder	Single component	Selective sintering	Heat	laser beam	Selective laser sintering (SLS)
	Coated powder	Selective sintering	Heat	laser beam	Selective laser sintering (SLS)
	One component and one binder	Selective powder binding	Chemical bond	Fine droplet beam of binder liquid	3D-printing or, MIT process, or, selective powder binding (SPB)

Advantages

➤ Design

- Other manufacturing processes require a careful and detailed analysis of the part geometry to determine things like the order in which different features can be fabricated, what tools and processes must be used, and what additional fixtures may be required to complete the part. In contrast,
- AM needs only some basic dimensional details and a small amount of understanding as to how the AM machine works and the materials that are used to build the part



Advantages

➤ Planning

- There is **no need to define a blank geometry** as no blanks are required. 2. Minimum operation or process planning. GMP's are based on one operation only. 3. No complicated scheduling and routing problem. The part is made in one set up.

➤ Tooling/Jig- Fixtures

- GMP's are **tool-less processes** and the whole complex tasks of tool selection and tool management are not needed. 2. **No clamping required**. Complex jigs and fixtures are also eliminated. 3. Mould dies and tool design, die design are eliminated.

➤ Automation

- Introduction of flexible automation is very convenient as the whole shape generation process is generalized in character. 2. The shape generation is done using a computer and interfacing the manufacturing unit with the whole system is very simple.

GMP : generative manufacturing practices